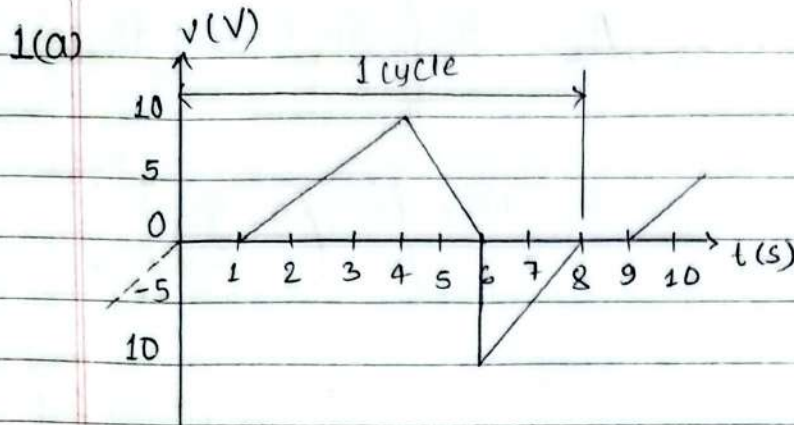




Soln:

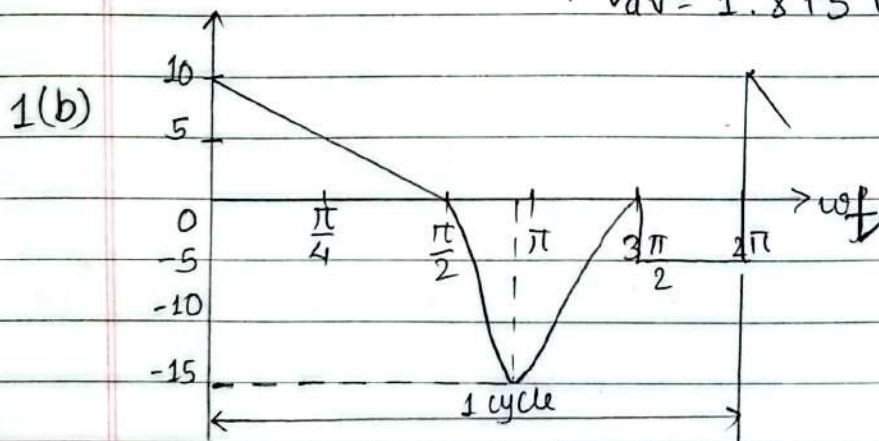
Average value (V_{av}) = $\frac{\text{Area enclosed by 1 complete cycle}}{\text{Total time period for 1-cycle}}$

$$= \frac{10 \times (4-1)}{2} + \frac{10(6-4)}{2} + \frac{-10(8-6)}{2}$$

$$= \frac{\frac{30}{2} + 10 - 10}{8}$$

$$= \frac{30}{16}$$

$$\therefore V_{av} = 1.875 \text{ V}$$



Soln:

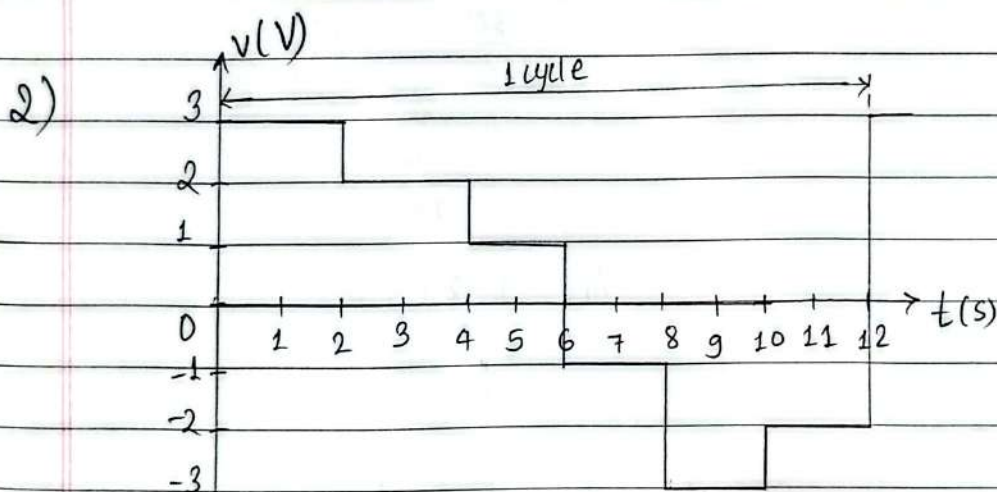
$$\text{Area of curve (from 0 to } \frac{\pi}{2}) A_1 = \left(\frac{\frac{\pi}{2} \times 10}{2} \right) = \frac{5\pi}{2}$$

$$\text{Area of curve (} \frac{3\pi}{2} \text{ to } 2\pi) A_2 = \left(\frac{3\pi}{2} + 2\pi \right) (-5) = -\frac{5\pi}{2}$$

$$\begin{aligned}
 \text{Area of curve } \left(\frac{\pi}{2} \text{ to } \frac{3\pi}{2}\right) A_3 &= -15 \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \sin\left(\theta - \frac{\pi}{2}\right) d\theta \\
 &= -15 \times \left[\cos\left(\theta - \frac{\pi}{2}\right) \right]_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \\
 &= 15 \times \left[\cos\left(\frac{3\pi}{2} - \frac{\pi}{2}\right) - \cos 0 \right]
 \end{aligned}$$

$$\therefore A_3 = -30$$

$$\begin{aligned}
 \text{Avg. value of } I &= \frac{A_1 + A_2 + A_3}{2\pi} = \frac{-30}{2\pi} \\
 &= -4.775 \text{ mA}
 \end{aligned}$$

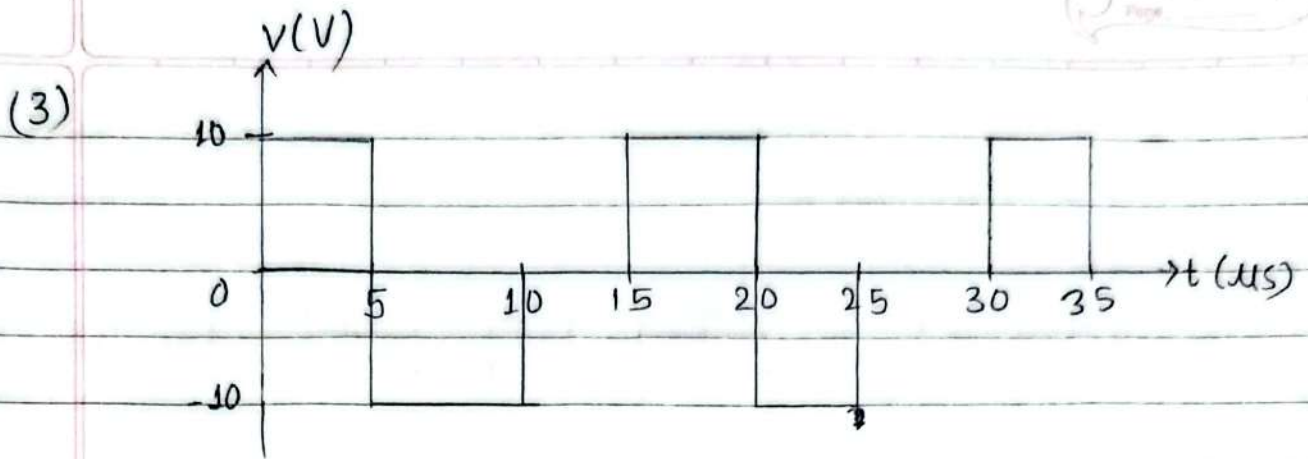


Solution:

$$V_{rms} = \sqrt{\frac{(3^2 \times 2) + (2^2 \times 2) + (1^2 \times 2) + ((-1)^2 \times 2) + ((-3)^2 \times 2) + ((-2)^2 \times 2)}{12}}$$

$$= \sqrt{4.667}$$

$$\therefore V_{rms} = 2.16 \text{ V}$$



Soln:

$$V_{av} = \frac{\text{Total sum (area)}}{\text{Total time}} = \frac{(5 \times 10) + (-10 \times 5) + (0 \times 5) + (10 \times 5) + (-10 \times 5) + (0 \times 5) + (10 \times 5)}{30}$$

$$= \frac{0}{30}$$

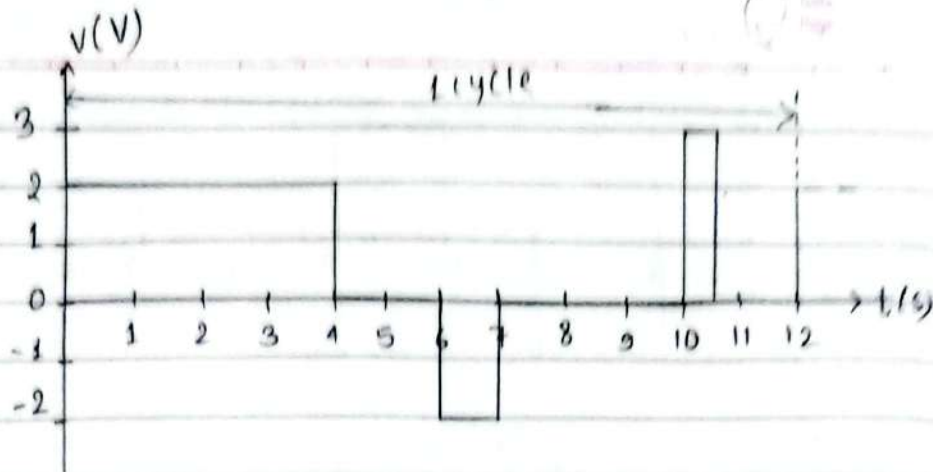
$$\therefore V_{av} = 0V$$

$$V_{rms} = \sqrt{\frac{(10)^2 \times 5 + 10^2 \times 5 + 0^2 \times 5}{15}}$$

$$= \sqrt{\frac{1000}{15}}$$

$$\therefore V_{rms} = 8.16V$$

(4)



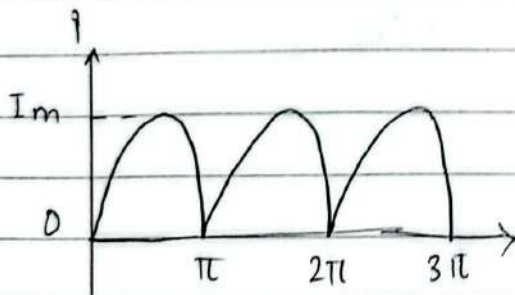
Soln:

$$V_{rms} = \sqrt{\frac{(2)^2 \times 4 + (0)^2 \times 2 + (-2)^2 \times 1 + (0)^2 \times 3 + (3)^2 \times 0.5 + (0)^2 \times 2}{12}}$$

$$= \sqrt{\frac{49}{24}}$$

$$\therefore V_{rms} = 1.429 \text{ V}$$

(5)



Here,

For 0 to π

$$A = \int_0^{\pi} I_m \sin \theta d\theta$$

$$= \int_0^{\pi/2} I_m \sin \theta d\theta + \int_{\pi/2}^{\pi} -I_m \sin \theta d\theta$$

$$= I_m + I_m$$

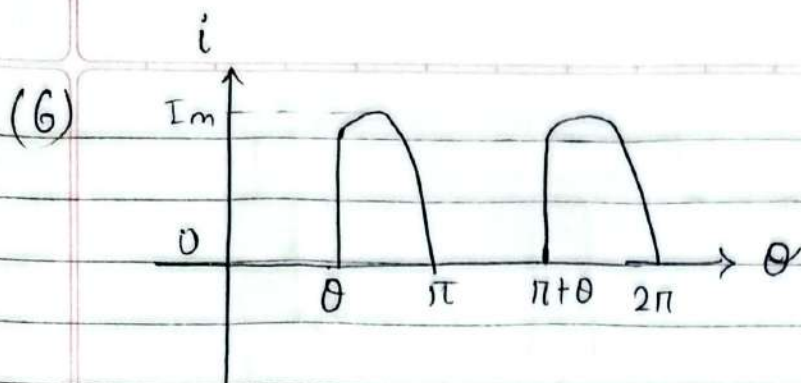
$$= 2I_m$$

Again,

$$\text{Avg. value } (V_{av}) = \frac{\left(\frac{2I_m}{2}\right)}{\left(\frac{\pi}{2}\right)} = \frac{2I_m}{\pi}$$

Again,

$$V_{rms} = \frac{I_m}{\sqrt{2}} \quad \left[\because \text{Peak} = \sqrt{2} V_{rms} \right]$$



Soln:

From 0 to θ $i(\alpha) = 0$

From θ to π $i(\alpha) = I_m \sin \alpha$

Now,

$$I_{avg} = \frac{1}{\pi} \int_0^{\pi} i(\alpha) d\alpha = \frac{1}{\pi} \left[\int_{\theta}^{\pi} I_m \sin \alpha d\alpha + \int_0^{\theta} 0 d\alpha \right]$$

$$= \frac{I_m}{\pi} \int_{\theta}^{\pi} \sin \alpha d\alpha$$

$$= \frac{I_m}{\pi} [-\cos \alpha]_{\theta}^{\pi}$$

$$= \frac{I_m}{\pi} [\cos \theta - \cos \pi]$$

$$= \frac{I_m}{\pi} [1 + \cos \theta]$$

we have,

$$I_{av} = \frac{1}{2} I_m$$

$$\frac{I_m}{\pi} [1 + \cos \theta] = \frac{I_m}{2}$$

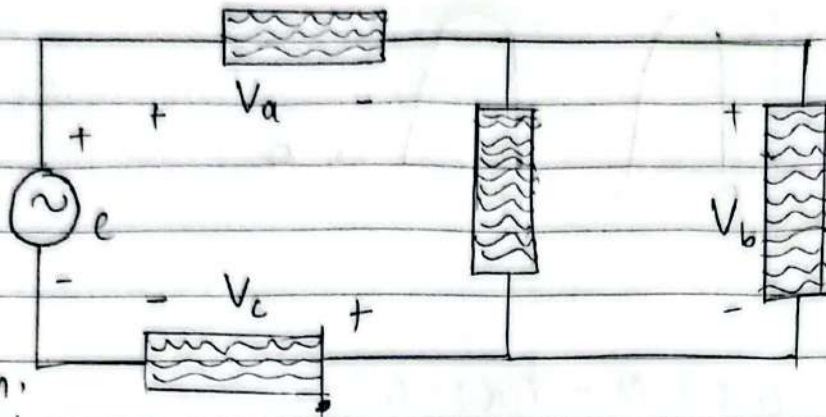
$$2 + 2 \cos \theta = \pi$$

$$\cos \theta = \frac{\pi - 1}{2}$$

$$\theta = \cos^{-1} \left(\frac{\pi - 1}{2} \right)$$

$$\therefore \theta = 55.19^\circ$$

7)

Soln:

$$V_a = 60 \sin \omega t = \left(\frac{60}{\sqrt{2}} \right) \angle 0 = 42.43 \angle 0$$

$$V_b = 30 \sin \omega t = \left(\frac{30}{\sqrt{2}} \right) \angle 0 = 21.21 \angle 0$$

$$e = 120 \sin (\omega t + 30) = \left(\frac{120}{\sqrt{2}} \right) \angle 30 = 60\sqrt{2} \angle 30$$

$$V_c = ?$$

Now,

$$60\sqrt{2} \angle 30 - (42.43 \angle 0) - 21.21 \angle 0 = V_c$$

$$\therefore V_c = 43.554 \angle 76.94$$

$$\begin{aligned} \therefore V_c &= 43.554 \times \sqrt{2} \sin(\omega t + 76.94^\circ) \\ &= 61.59 \sin(\omega t + 76.94^\circ) \end{aligned}$$